

Real-Time Group Chat Application

Computer System Design (CSD) — Assignment 4 | Persistence, Security & Verification Report

Course: Computer System Design (CSD)	Assignment: Assignment 4
Host SSH Machine: student@10.1.75.51 (Port 2237)	Internal Port: 5000
Official Testing Public URL: http://10.1.75.51:5237/	Tech Stack: Flask, Flask-SocketIO, SQLite, AES-256-GCM, Ed25519

1. Executive Summary & Lab Network Mapping

This project implements a multi-user, real-time Group Chat Application for **Computer System Design (CSD) Assignment 4**. Built using WebSockets (Flask-SocketIO) and SQLite persistence, it includes advanced security layers (AES-256-GCM confidentiality and Ed25519 digital signatures). The backend server runs on the allotted SSH laboratory machine (student@10.1.75.51).

Lab Network & Port Forwarding Mapping:

The Flask backend process binds internally to port 5000. The laboratory network NAT router maps SSH port 2237 to external web port 5237. Therefore, the official public URL for TA evaluation across all 4 student clients is: <http://10.1.75.51:5237/>.

2. Team Members & Key Technical Contributions

Role / Designation	Student Name	Roll No.	SSH Server	Key Technical Contribution
Group Head (Host)	Ranga Chandra Naga Venkata Chaitanya	12341740	ssh -p 2237	Backend Architecture, Flask-SocketIO Core, AES-256-GCM Encryption
Member 2	Bhukya Raju	12340520	ssh -p 2238	Front-End UI Design, Cyberpunk Aurora Glassmorphism Theme, SocketIO Client
Member 3	V.G.N. Harshitha	12342310	ssh -p 2239	Database Schema Architecture (chat.db), SQLite Message Persistence
Member 4	Maloth Madhu	12341370	ssh -p 2240	Multi-Client Integration & Testing across SSH Machines (10.1.75.51)

3. Demonstrations: Persistence, Tamper Detection & Signatures

■ Screenshot 1: Real-Time Multi-User Chat UI Demonstration (http://10.1.75.51:5237/) [Paste Screenshot Here]
■ Screenshot 2: Demonstration of Message Persistence (Server Process Restart & Auto-History Load) [Paste Screenshot Here]
■ Screenshot 3: Demonstration of Tamper Detection (AES-256-GCM Auth Tag Failure via demo_tamper.py) [Paste Screenshot Here]
■ Screenshot 4: Demonstration of Signature Verification (Ed25519 Public Keys & Signatures in sqlite3) [Paste Screenshot Here]

4. Database Verification Commands (SSH Host)

To inspect encrypted messages and signatures in SQLite:
sqlite3 ~/chat_app/chat.db "SELECT id, sender, ciphertext, signature, timestamp FROM messages;"

To verify public Ed25519 user signing keys:
sqlite3 ~/chat_app/chat.db "SELECT username, public_key FROM signing_keys;"

To trigger Tamper Detection test:
python3 ~/chat_app/demo_tamper.py

5. Official Submission Details

- **GitHub Repository:** <https://github.com/chaitanyakumarAI/Group-Chat>
- **Live Application URL (For TA Testing):** <http://10.1.75.51:5237/>
- **Host SSH Server Command:** `ssh -p 2237 student@10.1.75.51`
- **Daemon Process Command:** `nohup python3 server.py > server.log 2>&1 &`